1 Solution — GIAM 1.1 Problem 6

1.1 Problem

Find the first 20 decimal places of π, $3/7$, $\sqrt{2}$, $2/5$, $16/17$, $\sqrt{3}$, $1/2$, and $42/100$. Classify each as **terminating**, **having a repeating pattern**, or **showing no discernible pattern**.

1.2 Computation

For the irrationals $\pi, \sqrt{2}, \sqrt{3}$: use standard high-precision tables (or any calculator that handles 20+ digits). For the rationals: long-divide until either a 0 remainder appears (terminating) or until a remainder *repeats* (periodic — the period is then the cycle length).

The eight expansions to the first 20 decimal places:

- $\pi \quad = \; 3.14159\;26535\;89793\;23846\;\ldots$
- $3/7 \quad = \; 0.42857\;14285\;71428\;57142\;\ldots$
- $\sqrt{2} \quad = \; 1.41421\;35623\;73095\;04880\;\ldots$
- $2/5 \quad = \; 0.40000\;00000\;00000\;00000\;\ldots$
- $16/17 \;\; = \; 0.94117\;64705\;88235\;29411\;\ldots$
- $\sqrt{3} \quad = \; 1.73205\;08075\;68877\;29352\;\ldots$
- $1/2 \quad = \; 0.50000\;00000\;00000\;00000\;\ldots$
- $42/100 \; = \; 0.42000\;00000\;00000\;00000\;\ldots$

1.3 Visualizing the Three Categories

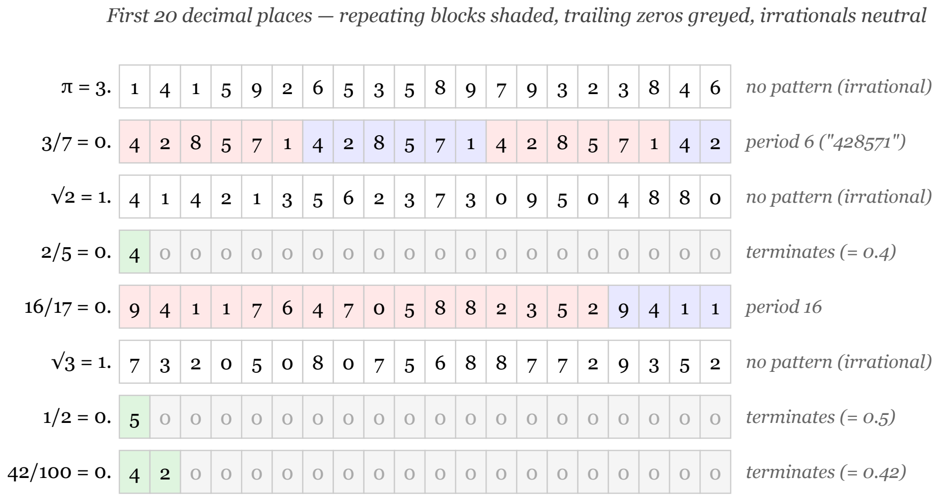

Figure 1.1: Each row shows the first 20 decimal digits. Pink/blue shading marks the alternating periods for repeating expansions; grey trailing zeros for terminating ones; plain white for the no-pattern irrationals.

1.4 Classification

Quantity	**First 20 decimal places**	**Classification**	**Reason**
π	$3.14159\,26535\,89793\,23846\ldots$	no discernible pattern	π is irrational (classical theorem).
$3/7$	$0.42857\,14285\,71428\,57142\ldots$	period 6 ($\overline{428571}$)	Rational; denominator 7 has prime factor $\notin \{2, 5\}$.
$\sqrt{2}$	$1.41421\,35623\,73095\,04880\ldots$	no discernible pattern	$\sqrt{2}$ is irrational.
$2/5$	$0.40000\,00000\,00000\,00000\ldots$	terminating (0.4)	Rational; denominator $5 = 5^1$ has only prime 5.

$16/17$	$0.94117\,64705\,88235\,29411\ldots$	period 16	Rational; denominator 17 is prime $\notin \{2,5\}$; period $= \mathrm{ord}_{17}(10) = 16$.
$\sqrt{3}$	$1.73205\,08075\,68877\,29352\ldots$	no discernible pattern	$\sqrt{3}$ is irrational.
$1/2$	$0.50000\,00000\,00000\,00000\ldots$	terminating (0.5)	Rational; denominator $2 = 2^1$ has only prime 2.
$42/100$	$0.42000\,00000\,00000\,00000\ldots$	terminating (0.42)	$42/100 = 21/50$; denominator $50 = 2 \cdot 5^2$ has only primes $2, 5$.

Table 1.1

1.5 Reading the Table — Three Categories

- **Terminating** ($2/5$, $1/2$, $42/100$) — the denominator (in lowest terms) factors as $2^m 5^n$, so the long-division remainder reaches 0 after finitely many steps. From that point on the expansion is just trailing zeros. This is the Problem 5 characterisation in action.
- **Repeating** ($3/7$, $16/17$) — the denominator has at least one prime factor outside $\{2,5\}$, so long-division can never zero its remainder. By pigeonhole the remainders must cycle, and once they cycle the digits do too. The period divides $\varphi(\text{denom})$.
- **No pattern** (π, $\sqrt{2}$, $\sqrt{3}$) — these are *irrational*, so by Problem 3's principle their decimal expansions are not eventually periodic. A finite truncation looks "random."

1.6 Remark — Why the Period Lengths Are What They Are

For a rational a/b in lowest terms with $\gcd(b,10) = 1$, the period of the repeating decimal equals the **multiplicative order of 10 modulo b** — that is, the smallest positive integer k such that $10^k \equiv 1 \pmod{b}$.

- $b = 7$: $10^1 \equiv 3$, $10^2 \equiv 2$, $10^3 \equiv 6$, $10^4 \equiv 4$, $10^5 \equiv 5$, $10^6 \equiv 1 \pmod{7}$. Period $= 6$. ✓

- $b = 17$: working through the powers, $10^{16} \equiv 1 \pmod{17}$ is the first time. Period $=$ 16. ✓ (This is in fact the full multiplicative order — 17 is a "long-period prime" for base 10.)

By Fermat's little theorem, the period always *divides* $\varphi(b) = b - 1$ when b is prime. For $b = 7$: divides 6, equals 6. For $b = 17$: divides 16, equals 16.

1.7 Remark — A Trick for 16/17 Without Long Division

The trick: write $1/17 = 0.\overline{0588235294117647}$ (period-16 expansion of the reciprocal), then $16/17 = 1 - 1/17 = 0.\overline{9411764705882352}$. Subtracting from 1 flips each digit "9 → 0, 4 → 5, 1 → 8, ..." — equivalently $d \mapsto 9 - d$ — and inserts a final "carry" that shifts the periodic block by one position. The result is itself a period-16 expansion, just with digits complemented.

Concretely: $1/17 = 0.\overline{0588235294117647}$; flipping each digit gives $0.\overline{9411764705882352}$, which is $16/17$. The pattern matches our long-division answer exactly.

1.8 Remark — 42/100 vs 21/50: Same Number, Same Classification

Both $42/100$ and $21/50$ are valid representations of the same rational 0.42. The "lowest-terms" version is $21/50$:

$$42/100 \;=\; \frac{2 \cdot 21}{2 \cdot 50} \;=\; \frac{21}{50}.$$

The denominator-prime-factor rule from Problem 5 looks at the *lowest-terms* denominator: here that's $50 = 2 \cdot 5^2$, whose only primes are 2 and 5 — hence terminating. Even if we use the un-reduced form $42/100$, the denominator $100 = 2^2 \cdot 5^2$ has the same primes; same conclusion. The classification doesn't depend on whether the fraction is reduced.

1.9 Remark — The Irrationality of $\sqrt{2}, \sqrt{3}$ Is Elementary

Unlike π, the irrationality of $\sqrt{n}$ (for n a positive non-square integer) has a one-line proof. If $\sqrt{n} = a/b$ in lowest terms with $b > 1$, then $n = a^2/b^2$, so $nb^2 = a^2$. Considering any prime $p \mid b$: $p \mid a^2$, so $p \mid a$, contradicting $\gcd(a, b) = 1$. Hence $b = 1$, and $n = a^2$ is a perfect square — contradicting our hypothesis. ■

Applied to $n = 2$ or $n = 3$: 2 and 3 are not perfect squares, so $\sqrt{2}$ and $\sqrt{3}$ are irrational. Their decimal expansions are not eventually periodic.

1.10 Remark — Why π Is the Famous Hard Case

The "no discernible pattern" verdict for π is right, but proving it requires *non-elementary* tools (calculus integrals à la Niven, or complex analysis via Lindemann). For the purposes of Chapter 1 it's an accepted fact, with the understanding that a proper proof comes much later in a typical analysis course.

The visual evidence in the figure — 20 digits with no obvious repetition — is suggestive but not a proof. (After all, $16/17$'s period is also 16 digits, and on a glance of 20 places, you might miss the cycle.) The actual proof has to *guarantee* no cycle ever appears.

1.11 Remark — How to Spot a Long-Period Rational

A common pitfall: with $16/17$'s 16-digit period, the first 20 decimal places contain only *one full period plus a 4-digit start of the next.* The cycle isn't visually obvious until you write out, say, 40 digits:

$$16/17 \;=\; 0.\underbrace{9411764705882352}_{\text{period 1}}\,\underbrace{9411764705882352}_{\text{period 2}}\,9411\,\ldots$$

To distinguish a long-period rational from an irrational with just 20 places of decimal output, you need either (i) more decimal places, or (ii) the knowledge that the denominator is a particular prime and the period theoretically could be that long. Decimal-expansion *inspection alone* is a poor diagnostic for rationality past period 20 or so.

This is one reason mathematicians prefer the prime-factor characterisation (Problem 5) over a "look at the decimals" approach: it's a *finite, decidable* test, even when the period would be impractically long to verify by inspection.